

Topic 10: Course Conclusion

Spectral methods, optional projects and research at the MTG

Audio Signal Processing for Music Applications

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Concluding class roadmap

A review of what the methods enable, where they fail and how to extend them

1 Course review

Spectra, models, transformations and descriptions

2 Open problems

Time resolution, transients, tracking and context

3 MTG examples

Historical applications and current research

4 Creative lab

A musical piece made with sound transformations

5 Analysis labs

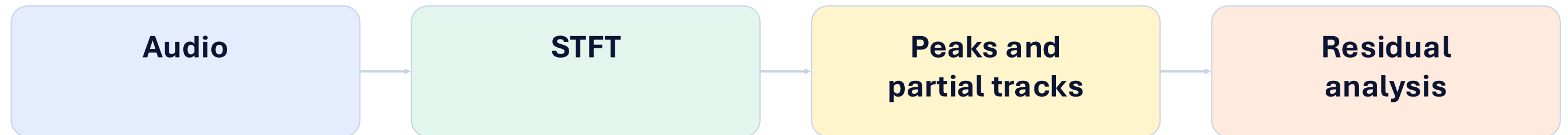
Richer features and multiresolution models

6 Next steps

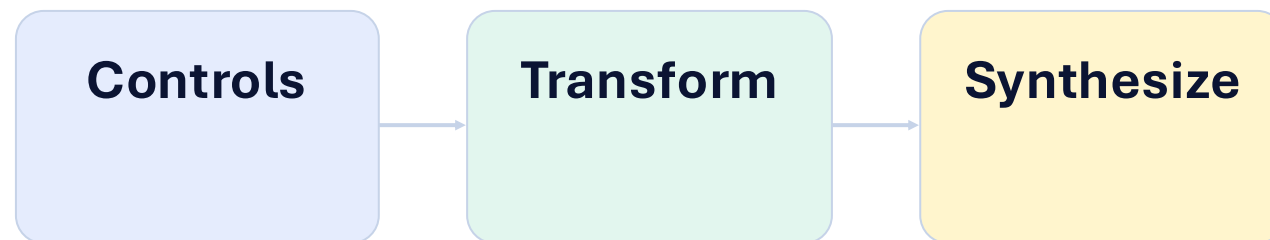
A well-defined question with convincing evidence

The course framework

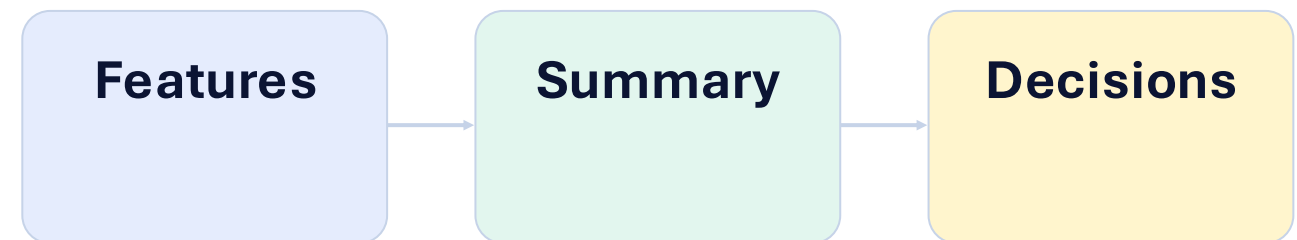
The same analysis can support sound processing and sound description



Sound processing



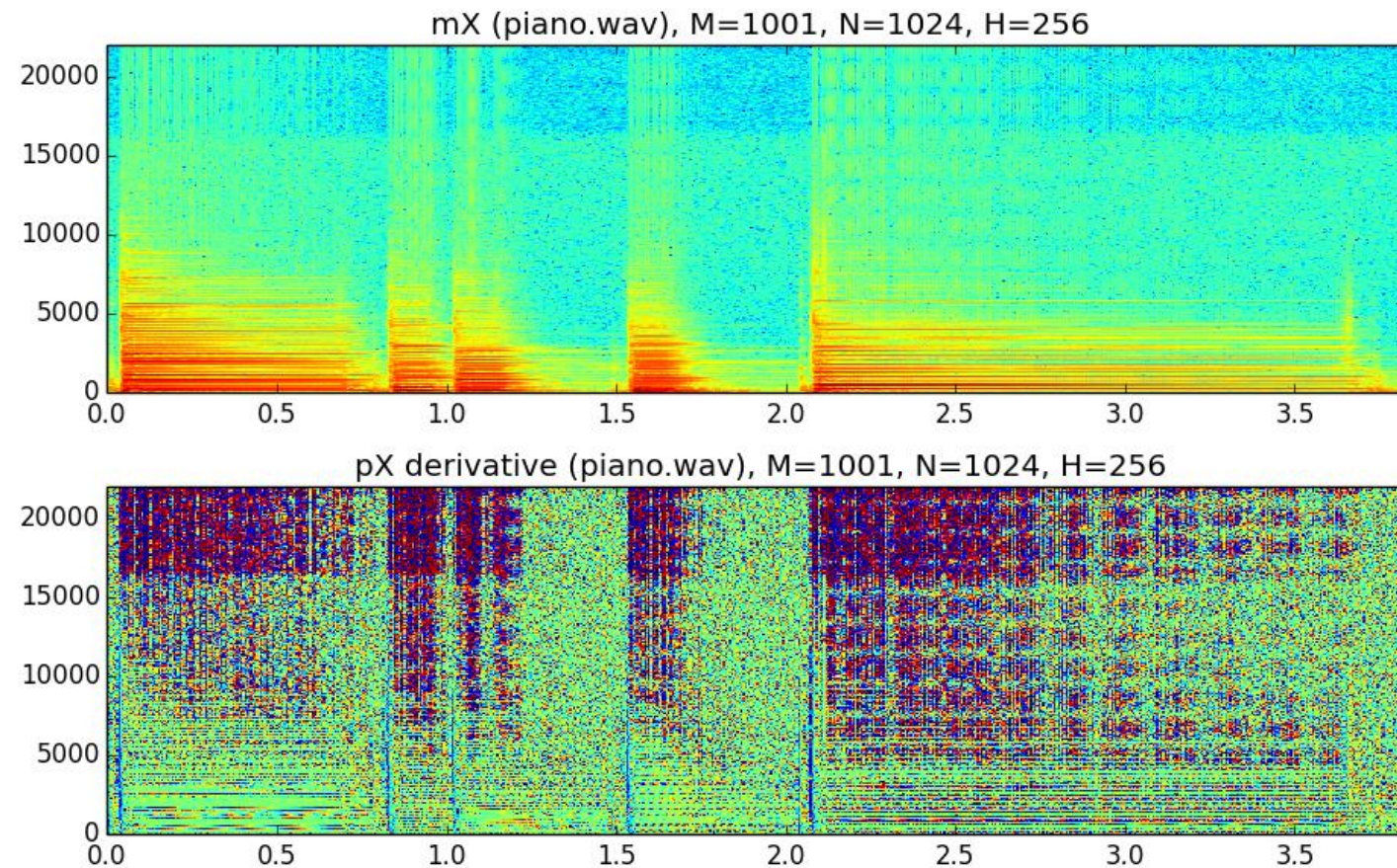
Sound description



A useful representation exposes what the application needs to control or measure.

A spectral view of musical sound

A spectrogram reveals how energy is distributed across frequency and time



Original 10T2 piano example: magnitude and phase derivative

Reading the plots

- Horizontal bands suggest sustained partials.
- Vertical changes often mark attacks.
- The lower plot shows phase variation.
- Phase is sensitive where magnitude is weak.

DFT and STFT

The DFT analyses a finite sequence, and the STFT repeats a windowed analysis

Discrete Fourier transform

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{\frac{-j 2\pi k n}{N}}$$

Short-time Fourier transform

$$X_{\ell}[k] = \sum_{n=0}^{M-1} w[n] x[n + \ell H] e^{\frac{-j 2\pi k n}{N}}$$

M: window length N: FFT size, $N \geq M$ H: hop in samples
 $k = 0, \dots, N-1$ ℓ : frame index

Analysis parameters and resolution

Window length, FFT size and hop size control different aspects of the analysis

Parameter	Main effect	Limitation
Window length M	Longer support separates nearby stable components more easily	Fast changes spread across a longer interval
Window type	Sets main-lobe width and sidelobe leakage	Better sidelobe suppression often widens the main lobe
FFT size N	Grid spacing is f_s/N	Zero-padding samples the spectrum more densely
Hop size H	Time between frames is H/f_s	Smaller hops add frames without shortening the window

Peak sharpness alone does not establish finer resolving power.

Sinusoidal and harmonic models

A sinusoidal representation separates amplitude, frequency and phase trajectories

Sinusoidal component

$$s[n] = \int_{r=1}^R A_r[n] \cos(\varphi_r[n])$$

$$\varphi_r[n+1] = \varphi_r[n] + \frac{2\pi f_r[n]}{f_s}$$

Frequency f is in Hz. Phase advances in radians.
Initial phases and continuity affect the waveform.

Harmonic constraint

$$f_r[n] \approx r f_0[n]$$

- The harmonic model ties partials to a fundamental.
- A free sinusoidal model also represents inharmonic partials.
- A single f_0 cannot describe several independent pitched sources.

Residual and stochastic components

The remainder contains everything the sinusoidal component did not explain

$$r[n] = x[n] - s[n]$$

$$\hat{x}[n] = s[n] + \eta[n]$$

Representation	What it preserves	Typical use
Residual waveform $r[n]$	Timing and phase of the remaining signal	Detailed resynthesis, including unmodelled events
Stochastic component $\eta[n]$	A noise-like signal shaped by a spectral envelope	Compact control of noise texture

Transformations in a model space

Independent controls are useful only when the representation separates the relevant properties

Control	Musical effect	What needs attention
Partial frequencies	Pitch shift or harmonic stretching	Phase continuity, aliasing and spectral envelope
Amplitudes and envelope	Brightness, balance and dynamics	Timbre can change even when f_0 stays fixed
Time trajectories	Time stretching and event rearrangement	Attacks, track interpolation and new synthesis timing
Stochastic envelope	Noise colour and texture	A noise model may blur structured residual events
Morphing parameters	Interpolation between analysed sounds	Correspondence between components and events

Sound description and musical interpretation

Higher-level concepts need temporal relationships, models and listening context

Level	Examples	Evidence needed
Physical	Spectrum, frequency, intensity, onset time	Signal measurements
Sensorial	Pitch, loudness, timbre	Perceptual relationships to those measurements
Perceptual	Intervals, beat, dynamics	Relationships between components and events
Formal	Melody, key, meter, instrumentation	Musical models and temporal organisation
Cognitive	Style, emotion, semantic concepts	Listeners, cultural context and task definitions

What a model leaves unexplained

A reconstruction error can reveal a mismatch between the sound and the assumptions

Common mismatches

- Sinusoidal peaks can miss brief attacks or diffuse noise.
- A single harmonic series can fail on polyphonic mixtures.
- A global feature mean hides event order and local changes.

Useful checks

- Inspect partial continuity, residuals and spectrograms.
- Compare level-matched original and baseline resynthesis.
- Test another model or one parameter change and explain the difference.

Audio processing beyond this course

The next advances address uncertainty, continuity and components with different time scales

Analysis extensions

- Better peak estimation and partial tracking under overlap.
- Explicit transient modelling and adaptive or multiresolution windows.
- Residual models that separate noise from structured events.

Modelling extensions

- More robust phase-aware synthesis and parameter interpolation.
- Physical models that represent sound-production mechanisms.
- Learned models with data-dependent capabilities and limitations.

Music information beyond the audio waveform

Different sources describe different aspects of the same musical activity

Information source	What it can add	Processing approaches
Scores and MIDI	Intended notes, timing and musical structure	Symbolic pattern analysis and score alignment
Lyrics and text	Language, descriptions and documentary context	Language models and semantic analysis
Gestures and video	Performance actions and visual interaction	Motion analysis, computer vision and multimodal learning
Metadata and communities	Artists, relationships, traditions and uses	Knowledge graphs, ontologies and network analysis
Listeners and performers	Experience, usability and cultural meaning	Perception studies and human-computer interaction

Evidence for a useful music technology

An evaluation should measure the property that the application claims to improve

Technical evidence

- Use a clear baseline and change one design choice at a time.
- Keep datasets, analysis settings and evaluation procedures comparable.
- Report variability and inspect difficult or failed cases.

Musical evidence

- Listen for the intended change and unwanted artefacts.
- Use musical context to interpret similar feature values.
- Ask performers or listeners whether the result supports the intended task.

The course within MTG research

The MTG combines signal processing with description, interaction and shared resources

Research area	Course foundations	Examples in this class
Audio signal processing	Spectral models, residuals, synthesis and transformation	Singing synthesis, voice transformation, source separation
Sound and music description	Features, pitch, chroma, onsets and similarity	Essentia, CompMusic and learned representations
Musical interaction	Real-time control and meaningful parameters	Reactable and GrooveTransformer
Sound and music resources	Metadata, reproducible analysis and collections	Freesound, AcousticBrainz and open datasets

Music Technology Group, Department of Engineering, Universitat Pompeu Fabra

Singing synthesis: spectral models and Vocaloid

MTG research with Yamaha connected recorded voice material to flexible musical control



Historical examples reproduced from 10T3



Course connection

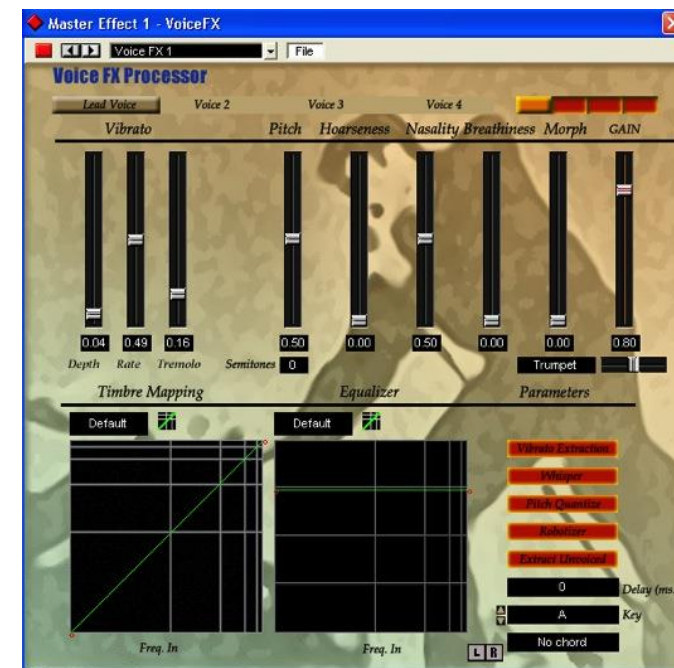
- Spectral processing adjusts pitch and timbre.
- Voice samples retain articulation detail.
- Expression also needs performance control.

Voice transformation: interpretable controls

Pitch, spectral shape and timing offer distinct ways to alter a voice



Historical MTG prototypes in 10T3

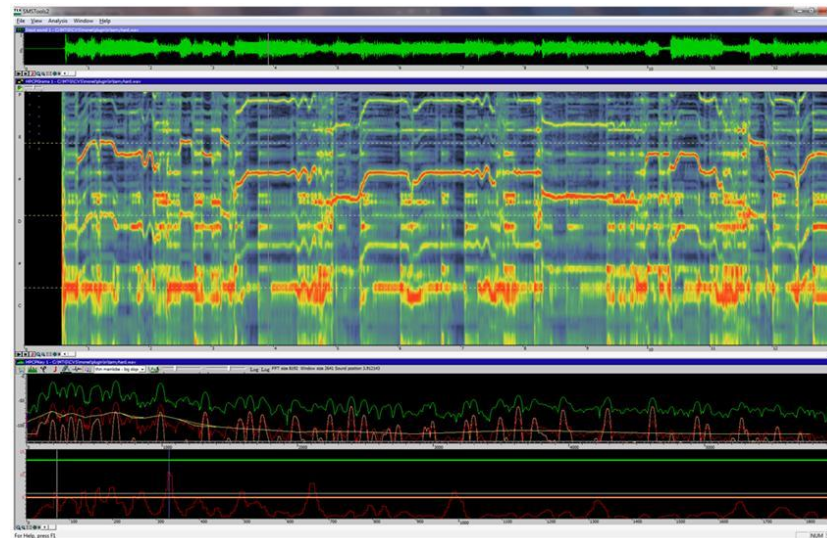
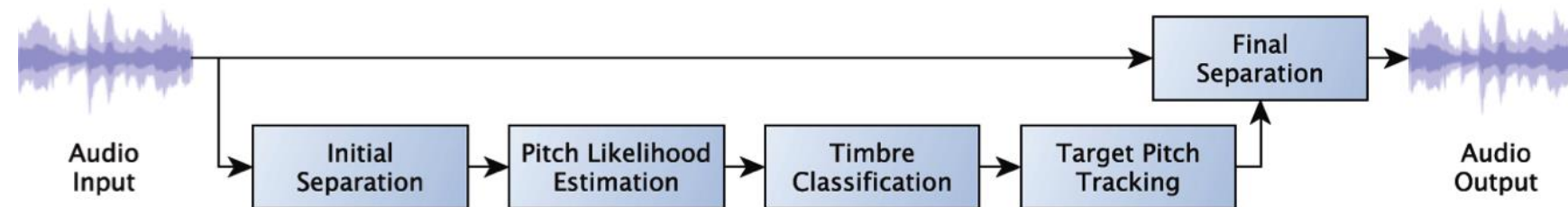


Course connection

- Frequency scaling changes pitch.
- Envelope changes reshape vocal timbre.
- Latency and continuity affect live use.

Source separation: overlapping musical components

Separating a singer from accompaniment requires evidence beyond isolated peak detection



Historical separation example from 10T3

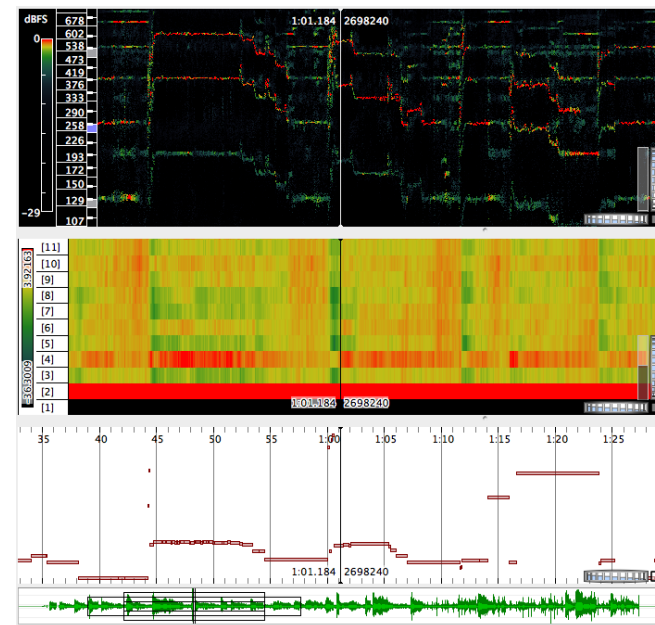
Past and recent work

- Pitch and timbre help assign components.
- Overlapping harmonics cause ambiguity.
- Carnatic vocal separation uses diffusion models in recent MTG work.

Essentia: reusable audio-analysis infrastructure

Course descriptors become components of larger, reproducible analysis pipelines

ESSENTIA



Essentia and feature analysis in 10T3

Course connection

- DSP blocks produce spectral and tonal features.
- Statistics support collection analysis.
- Deep-learning inference and API services extend the toolkit.

CompMusic and Dunya: culture-specific music description

Melody, rhythm and similarity depend on the musical tradition and the question



CompMusic, ERC project 2011–2017, and the Dunya collections

Course connection

- Pitch contours support melodic analysis.
- Rhythmic structure needs temporal models.
- Expert knowledge defines meaningful comparisons.

Reactable: sound processing in a musical interface

Tangible objects make synthesis and effects parameters available during performance



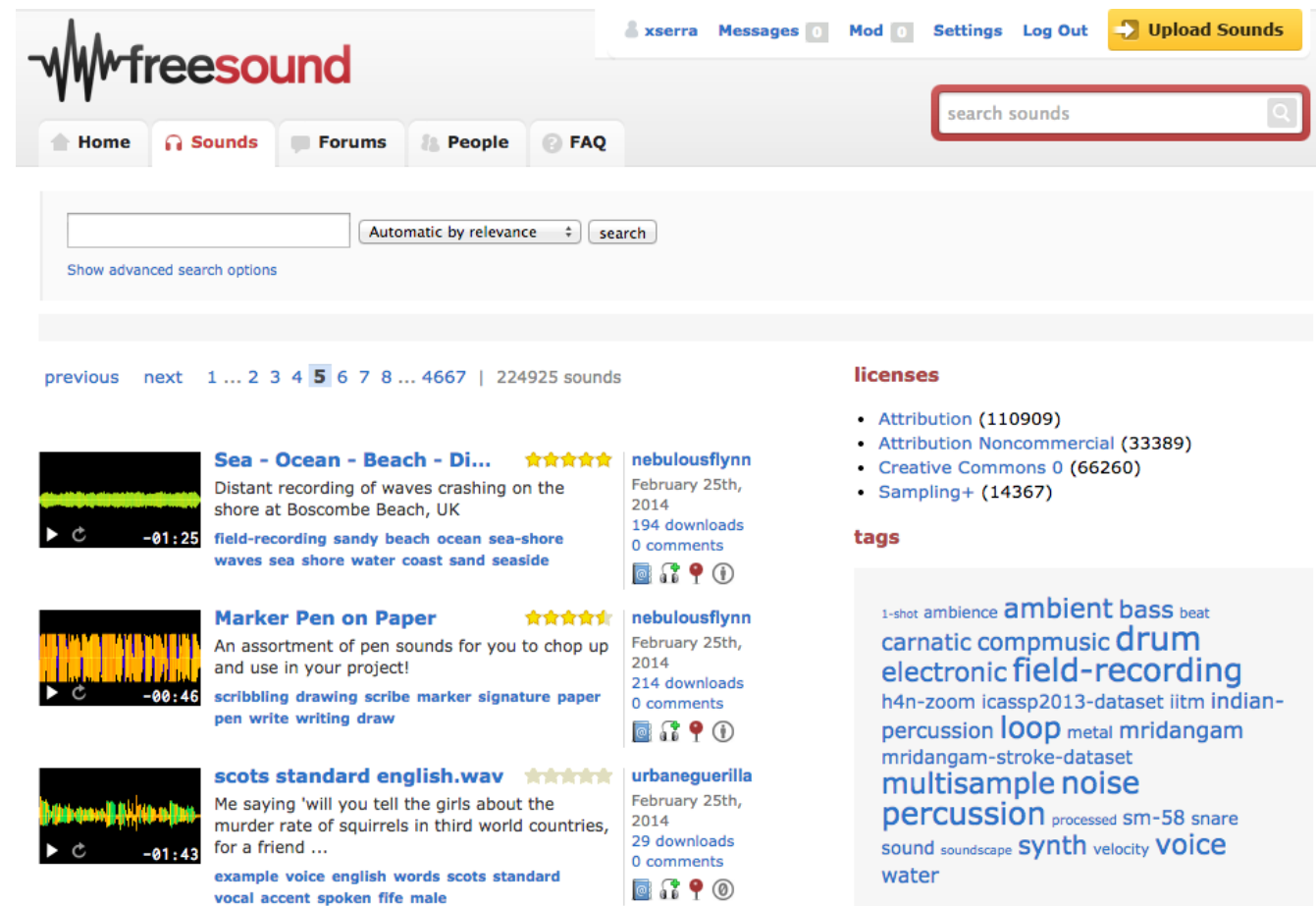
Historical Reactable performance image from 10T3

Course connection

- Objects control synthesis and effects.
- Audible and visual feedback guide the performer.
- Low latency and usable controls matter.

Freesound: audio, metadata and community

The course uses a research platform that joins shared sounds with descriptions and search



Historical Freesound interface in 10T3

Course connection

- Shared audio enables experiments and creative reuse.
- Features and tags support search.
- Dataset curation and licences affect reuse.

AcousticBrainz: a historical shared feature archive

A common extractor linked audio measurements to MusicBrainz recording identifiers

AcousticBrainz

[Contribute](#)[Download](#)[Goals](#)[API/Data](#)[FAQ](#)[Detailed Statistics](#)

Welcome to AcousticBrainz!

The AcousticBrainz project aims to crowd source acoustic information for all music in the world and to make it available to the public. This acoustic information describes the acoustic characteristics of music and includes low-level spectral information and information for genres, moods, keys, scales and much more. The goal of AcousticBrainz is to provide music technology researchers and open source hackers with a massive database of information about music. We hope that this database will spur the development of new music technology research and allow music hackers to create new and interesting recommendation engines.

AcousticBrainz is a joint effort between [Music Technology Group](#) at [Universitat Pompeu Fabra](#) in Barcelona and the [MusicBrainz](#) project. AcousticBrainz was originally envisioned by Xavier Serra, the founder and head of the MTG. At the heart of this project lies the [Essentia toolkit](#) from the MTG -- this open source toolkit enables the automatic analysis of music. The output from Essentia is collected by the AcousticBrainz project and made available to the public.

AcousticBrainz organizes the data on a track basis, indexed by the [MusicBrainz ID](#) for [recordings](#). If you know the MBID for a recording, you can easily fetch from AcousticBrainz. For details on how to do this, visit our [API documentation](#).

All of the data contained in AcousticBrainz is licensed under the [CC0 license](#) (public domain).

Examples

If you're wondering what this collected data actually looks like, have a look at the last 5 tracks that have been submitted:

1. [Nickelback / Just For](#)
2. [The National / Bloodbuzz Ohio](#)
3. [Lake of Tears / The Shadowshires](#)
4. [Cornershop / Breaking Every Rule Language English](#)
5. [Limp Bizkit / The Priest](#)

Track Statistics

	Lossy	Lossless	Total
Unique	388862	314427	703289
All	605298	399918	1005216

updated every 10 minutes
last updated: 2014-11-27 17:37
[more on the detailed statistics page](#)

Tweets



AcousticBrainz

@AcousticBrainz

2h

We have so many files our counter has overflowed its box! Thanks to everyone for 1m submissions and 700k uniques
[pic.twitter.com/SmNHHEock](#)

all music in the
scribes the acoustic
information for
is to provide music
se of information
w music technology
endation engines.
ersitat Pompeu
inally envisioned by

Track Statistics

	Lossy	Lossless	Total
Unique	388519	312628	701147
All	603604	396717	1000321

Historical AcousticBrainz interface. New data collection stopped in 2022.

Course connection

- Essentia produced low- and high-level features.
- Recording IDs linked features to metadata.
- Data collection ran from 2015 to 2022.

Audio Signal Processing for Music Applications - Course conclusion

MTG / UPF

23/44

Current MTG project: IMPA

Multimodal AI for Audio Processing, September 2024–August 2028

Research component	Connection to the course	Research extension
Dataset curation	Sound selection, preprocessing and annotations	Multimodal datasets for broader audio tasks
Audio representations	Spectra and selected descriptor vectors	Pre-trained representations learned from data
Task-specific models	Transformation, retrieval and classification	Models for generation, discovery and audio reuse
Evaluation and prototypes	Baselines, listening and failure analysis	Task-relevant metrics and application prototypes

Recent MTG research built on these foundations

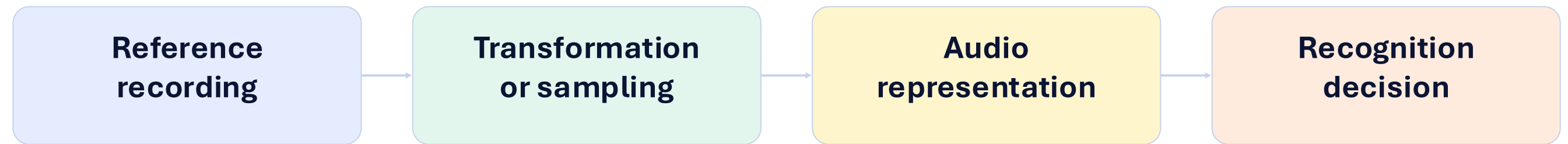
Different applications extend spectral analysis with learning, data and interaction

Example	Research contribution	Course connection
OMAR-RQ, 2025	Open self-supervised music-audio representations	Feature spaces, temporal representations and evaluation
Carnatic vocal separation, 2025	Regression-guided latent diffusion using live recordings	Mixtures, residuals and repertoire-specific evidence
GrooveTransformer, 2024	A generative drum-sequencing Eurorack module	Timing, musical controls and performance feedback

The DSP foundations remain useful for understanding inputs, controls, assumptions and errors.

Current MTG project: TROBA

Music recognition when recordings are transformed, sampled or regenerated



Connection to Topic 8 and the optional creative lab

- A useful recogniser retains identity through relevant changes.
- Pitch, timing, noise and excerpt length test different failure modes.
- A controlled transformation gives a reproducible robustness experiment.

The UPF–BMAT Chair on AI and Music

Music technology research also needs transparency, community participation and useful evaluation

Course practice	Broader research question	Concrete connection
Trace source sounds and settings	Can a result be inspected and attributed?	Provenance, openness and responsible reuse
Measure similarity and classify	What information does the representation privilege?	Distribution, recommendation and model evaluation
Transform or generate material	How can musicians direct and evaluate the result?	Creation and production tools with user participation
Explain results in musical context	Does a system support the intended learner or community?	Education, accessible interfaces and cultural context

Three optional labs

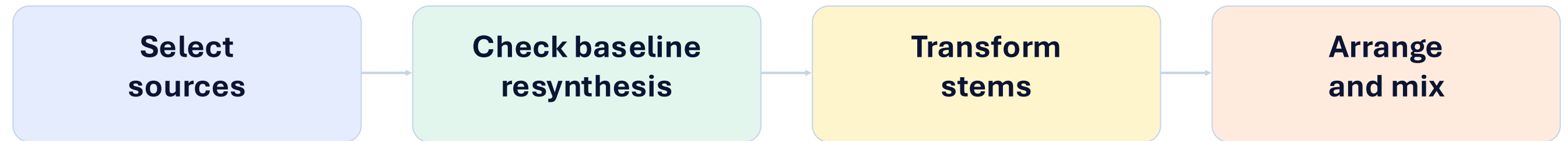
Each lab extends a different part of the course and can stand on its own

Optional lab	Main question	Expected result
E10-1: Music piece	How can spectral transformations serve a musical idea?	About one minute of original music with documented stems
E10-2: Music description	Do richer, cleaner features organise more instrument classes?	A controlled clustering comparison with Essentia
E10-3: Multiresolution	Can different windows improve the treatment of mixed content?	A sinusoidal implementation and comparisons on two sounds

Choose the question that interests you, then support your conclusions with listening and analysis.

Optional E10-1: a piece made with transformations

A musical goal determines the source material, model and processing chain



Greater freedom than E8

- Combine STFT, sinusoidal, residual and stochastic models.
- Process different sections differently and chain transformations when useful.
- Give each transformed stem a role, then build a coherent piece of about one minute.

E10-1: creative decisions and evidence

The final audio should make the purpose and consequence of each transformation audible

Before the final mix

- List Freesound links and justify model choices.
- Use two objective checks and one listening check for baseline quality.
- Keep one rejected model or failed transformation and explain the repair.

With the final piece

- Document each chain, key parameters and stem role.
- Describe entries, transitions, levels and masking choices.
- Explain one parameter change of $\pm 20\%$ and how it affects perception.

Optional E10-2: a larger instrument collection

A 10-class problem tests whether the descriptors capture more than a few easy contrasts

Collection and baseline

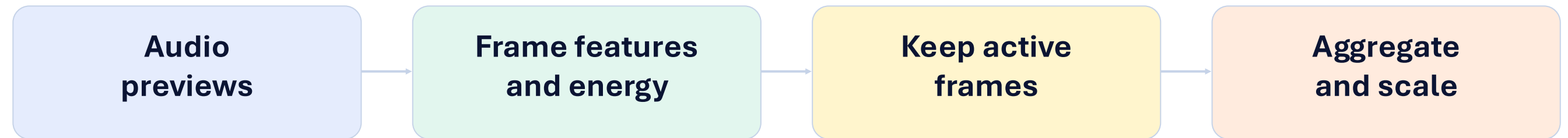
- Choose at least 10 instrument classes, with 20 coherent single notes or strokes per class.
- Use high-quality MP3 previews and keep sound IDs, links and query filters.
- Select descriptor subsets and run k-means with 10 clusters, repeated 10 times.

Questions to examine

- Which instrument pairs overlap?
- Does variation come from pitch, articulation or recording conditions?
- Are the same ambiguities stable across repeated runs?

E10-2: richer features and silent-frame removal

The experiment tests whether preprocessing and representation improve the chosen task



Two required improvements to investigate

- Compute additional spectral, temporal, tonal or MFCC descriptors with Essentia.
- Choose an energy threshold from example recordings and validate the retained frames.
- Compare the resulting groups using the same sounds and a consistent scaling rule.

E10-2: what the clustering score measures

The supplied majority-vote score is weighted cluster purity on the same collection

Experiment	Comparison purpose	Report over 10 runs
Freesound descriptors	Baseline organisation of the collection	Mean and standard deviation of purity
Essentia, all frames	Effect of the new extraction and feature set	Mean and standard deviation of purity
Essentia, active frames	Effect of removing silent frames	Mean and standard deviation of purity
Error inspection	Meaning of persistent confusions	Sound IDs, listening and descriptor evidence

10% is a uniform 10-class prediction baseline. It is not a chance baseline for this purity score.

Optional E10-3: multiresolution sinusoidal analysis

Different frequency bands can use different analysis durations

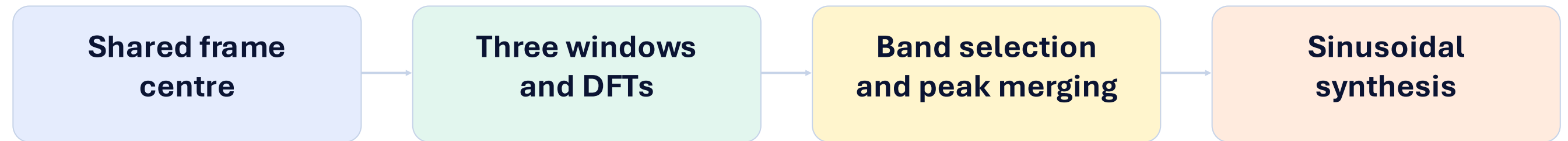
Band example	Window M / FFT N	Duration at 44.1 kHz	Intended emphasis
0–1000 Hz	4095 / 4096	92.9 ms	Stable low partials
1000–5000 Hz	2047 / 2048	46.4 ms	Intermediate tradeoff
5000 Hz–Nyquist	1023 / 1024	23.2 ms	Faster high-frequency changes

Suggested starting parameters from the lab. Tune the boundaries and windows for each sound.

A low-frequency attack still uses the long window in this fixed-band design.

E10-3: aligned analysis and consistent synthesis

All three analyses should describe the same frame centre before their peaks are merged



Implementation decisions

- Use frequency in Hz for band masks and normalise all windows consistently.
- Keep one hop, one time reference and compatible phase conventions across bands.
- Adapt the synthesis path of your installed sms-tools version and test overlap-add.

E10-3: the cost of several FFTs

The extra analysis cost depends on FFT sizes and how many frames are processed

Per-frame quantity	Single FFT	Three FFTs	Ratio
Full DFT bins	4096	$4096 + 2048 + 1024 = 7168$	1.75×
N log2 N proxy	49,152	$49,152 + 22,528 + 10,240 = 81,920$	1.67×
Frames per second	fs/H	fs/H if the hop is shared	Same frame count

These ratios concern analysis FFTs only. Peak tracking, synthesis, memory and hop size also affect runtime.

E10-3: evidence on two different recordings

Compare a tuned single-window baseline with a tuned multiresolution model

Experimental record

- Choose two distinct polyphonic recordings with melodic and percussive content, each no longer than 10 seconds.
- Record M, N, threshold, window, bands and hop for all four runs.
- Use the same display settings and level-matched listening for each comparison.

Interpretation

- Inspect onset smearing, close partials and band-boundary artefacts.
- Give independent settings for each sound and explain why they differ.
- Discuss cost and possible extensions to harmonic-plus-residual or stochastic models.

The optional labs as small research projects

Each activity isolates a decision that also appears in MTG applications

Optional lab	Course decision	MTG connection
Creative piece	Choice of representation and controlled transformation	Singing synthesis, voice transformation and TROBA robustness
Instrument description	Dataset curation, feature selection and aggregation	Essentia, Freesound and learned audio representations
Multiresolution analysis	Frequency-dependent support and synthesis consistency	Spectral-model research and representation of mixed musical content

A lab is a controlled experiment. A research application also needs scale, generalisation and user context.

Questions for a next project

A precise question makes modelling choices and evidence easier to justify

Research question	Controlled comparison	Evidence
Which transformations preserve recognisable source identity?	Vary one pitch, time or envelope control	Listening plus an identity/similarity task
Which descriptors organise a chosen instrument collection?	Same audio, feature ablations and preprocessing changes	Purity variability, confusions and inspected sounds
Which content benefits from multiresolution analysis?	Same recordings, tuned baselines and shared visual settings	Onsets, close partials, artefacts and processing cost

Further exploration at the MTG

Software, datasets and research training provide complementary next steps

Tools and materials

- sms-tools: inspect and extend the analysis/synthesis code.
- Essentia and Freesound: build analysis experiments and collections.
- MTG project pages and papers: connect methods to specific research questions.

Final takeaways

A model is useful when its assumptions fit the sound and the musical purpose

1 Representations

Spectra, tracks and features retain different information.

2 Resolution

Choose support and sampling for the changes you need to observe.

3 Control

Transform parameters while preserving continuity and useful structure.

4 Evidence

Check a baseline, listen and explain the errors.

5 Context

Musical meaning also needs people, data and cultural knowledge.

6 Research

A clear question and a reproducible comparison are a strong beginning.

References and credits

Sources for the review, optional projects and historical MTG examples

10T1: Beyond Audio Signal Processing, 10T2: Review, 10T3: MTG-UPF.

Optional notebooks: E10-1 Music piece, E10-2 Music description, E10-3 Multiresolution.

[sms-tools and the Audio Signal Processing for Music Applications materials.](#)

[Essentia algorithm documentation and the Essentia API project.](#)

[CompMusic / Dunya, Reactable, Freesound and AcousticBrainz.](#)

[Bonada and Serra, 2007: singing synthesis with performance sampling and spectral models.](#)

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